SOFTWARE REQUIREMENTS SPECIFICATION

(LOAD TRANSPORTATION ROBOTS)

CONTENTS

1. INTRODUCTION 3

1.1 PURPOSE 3

1.2 SCOPE 3

1.3 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS 4

1.4 REFERENCES 4

1.5 OVERVIEW 5

2. OVERALL DESCRIPTION 6

2.1 PRODUCT PERSPECTIVE 6

2.2 PRODUCT FUNCTIONS 6

2.2.1 Communicate with human 6

2.2.2 Communicate with other robots 6

2.2.3 Detect loads 6

2.2.4 Find passage 6

2.2.5 Monitor environment 6

2.3 USER CHARACTERISTICS 6

3. SPECIFIC REQUIREMENTS 7

3.1 SOFTWARE PRODUCT FEATURES 7

3.1.1 Communicate with human Use Case 8

3.1.1.1 Brief Description 8

3.1.1.2 Flow of Events 8

3.1.1.3 Special Requirements 8

3.1.1.4 Preconditions 8

3.1.1.5 Postconditions 8

3.1.1.6 Extension Points 8

3.1.1.7 Communicate with human Sequence Diagram 9

3.1.1.8 Communicate with human Activity Diagram 10

3.1.1.9 Communicate with human Statechart Diagram 11

3.1.2 Communicate with other robots Use Case 11

3.1.2.1 Brief Description 11

3.1.2.2 Flow of Events 11

3.1.2.3 Special Requirements 12

3.1.2.4 Preconditions 12

3.1.2.5 Postconditions 12

3.1.2.6 Extension Points 12

3.1.2.7 Communicate with other robots Sequence Diagram 12

3.1.2.8 Communicate with other robots Activity Diagram 13

3.1.2.9 Communicate with other robots State Chart Diagram 14

3.1.3 Detect loads Use Case 14

3.1.3.1 Brief Description 14
3.1.3.2 Flow of Events 14
3.1.3.3 Special Requirements 15
3.1.3.4 Preconditions 15
3.1.3.5 Postconditions 15
3.1.3.6 Extension Points 15
3.1.3.7 Detect loads Sequence Diagram 16
3.1.3.8 Detect loads Activity Diagram 17
3.1.3.9 Detect loads Statechart Diagram 18
3.1.4 Find passage Use Case 18
3.1.4.1 Brief Description 18
3.1.4.2 Flow of Events 18
3.1.4.3 Special Requirements 19
3.1.4.4 Preconditions 19
3.1.4.5 Postconditions 19
3.1.4.6 Extension Points 19
3.1.4.7 Detect loads Sequence Diagram 19
3.1.4.8 Find Passage Activity Diagram 20
3.1.4.9 Find Passage Statechart Diagram 21
3.1.5 Monitor environment Use Case 21
3.1.5.1 Brief Description 21
3.1.5.2 Flow of Events 21
3.1.5.3 Special Requirements 22
3.1.5.4 Preconditions 22
3.1.5.5 Postconditions 22
3.1.5.6 Extension Points 22
3.1.5.7 Monitor Environment Sequence Diagram 23
3.1.5.8 Monitor Environment Activity Diagram 24
3.1.5.9 Monitor Environment Statechart Diagram 26
3.2 NON FUNCTIONAL REQUIREMENTS 27
3.2.1 Usability 28
3.2.2 Safety 28
3.2.3 Performance 28
4. REQUIREMENTS TRACEABILITY 29

1. INTRODUCTION

1.1 Purpose

The purpose of this document is to highlight clearly and clarify the entire requirement on this load transportation robots represents in Unified Modeling Language (UML) which are use cases, class, state, activity and sequence diagram. The intended audience is Dr. Radziah, lecturer of Department of Software Engineering.

1.2 Scope

Load Transportation Robots as well as its name identified, will executes in order to help human bring a large object from one place to another place. The main purpose of the services is to find passage and existing through the passage while holding one large object on top of them. The system consists of two mobile robots where each robot is a differential drive wheeled mobile robot. The robots are leader and followers robot. Leader and followers strategy has been used where one mobile robot will be a leader while the other will be a follower. The robots are able to traverse in an environment and avoid obstacles (garbage) using sensor which is surrounded by four walls. The embedded controller is to move the wheels of the robot until the robot achieve its goal. In order to move the robot and achieve the goal, the software needs to control the motor at each robot drive wheels, monitor the environment, navigate the robot and cooperate with other robots. The robots also can interact to each other by sending or receiving messages known as robot conversation. The load detection is done via one push button. The controller communicates with other robots via a Radio Frequency (RF) transceiver and human through LCD and a switch. The robots must be activated by human at first by using a switch.

1.3 Definitions, Acronyms, and Abbreviations

The technical term involves is**:**

a) Actors which brought the meaning of role play and whom actually using the entire system internal and externally, direct or indirect to the system.

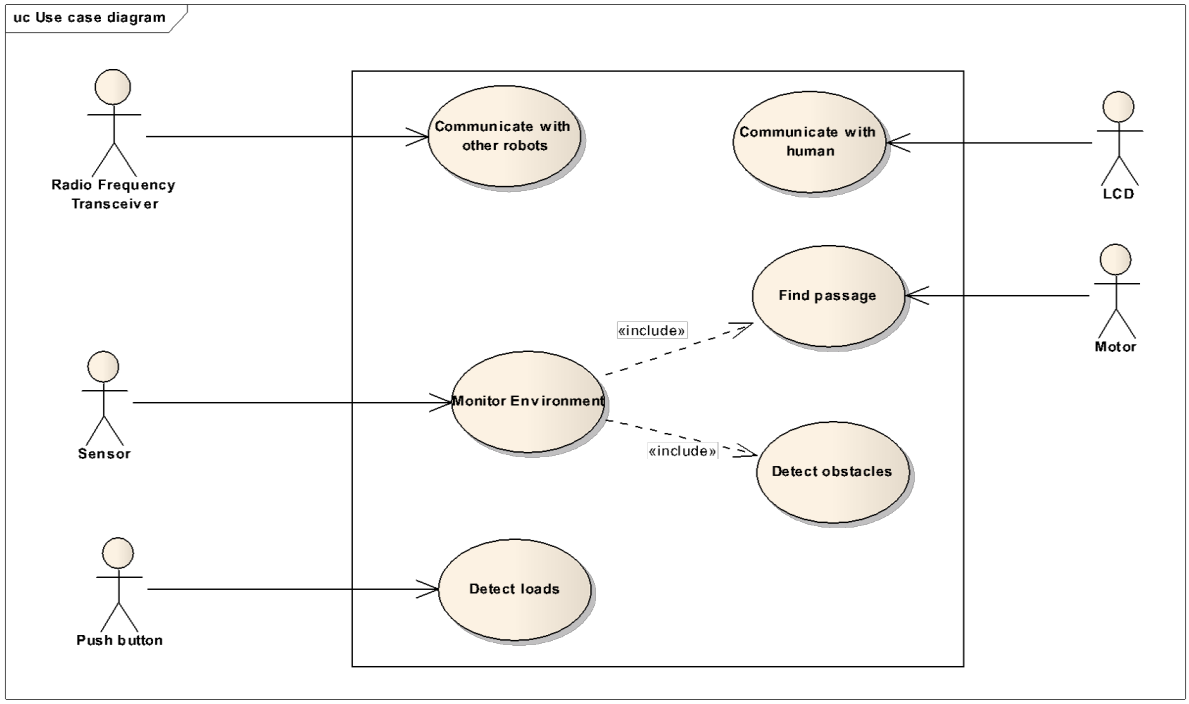
b) Functional Requirement meaning that the process of describing interaction between a system and its environment and how a system should behave under certain stimulation.

c) Non-Functional Requirement, it means the flow of describing restrictions on a system that limit the choices for its construction as a solution to the problem domain.

1.4 References

Miss Nor Baidura, research officer, Department of Software Engineering.

1.5 Overview


uc Use case diagram
Radio Frequency Transceiver
Communicate with other robots
Communicate with human
LCD
Find passage
Motor
«include»
Sensor
Monitor Environment
«include»
Detect obstacles
Push button
Detect loads

2. OVERALL DESCRIPTION

2.1 Product Perspective

This is a Stand-alone system.

2.2 Product Functions

2.2.1 Communicate with human

This use case is followed for the event of human communicates with the robot using LCD.

2.2.2 Communicate with other robots

This use case is followed for the event of the robot communicates with the other robots by sending message using Radio Frequency (RF) Transceiver.

2.2.3 Detect loads

This use case is followed for the event of human placing a large object on top of the robot.

2.2.4 Find passage

This use case is followed for the event of the robot moves to find a passage to deliver the load.

2.2.5 Monitor environment

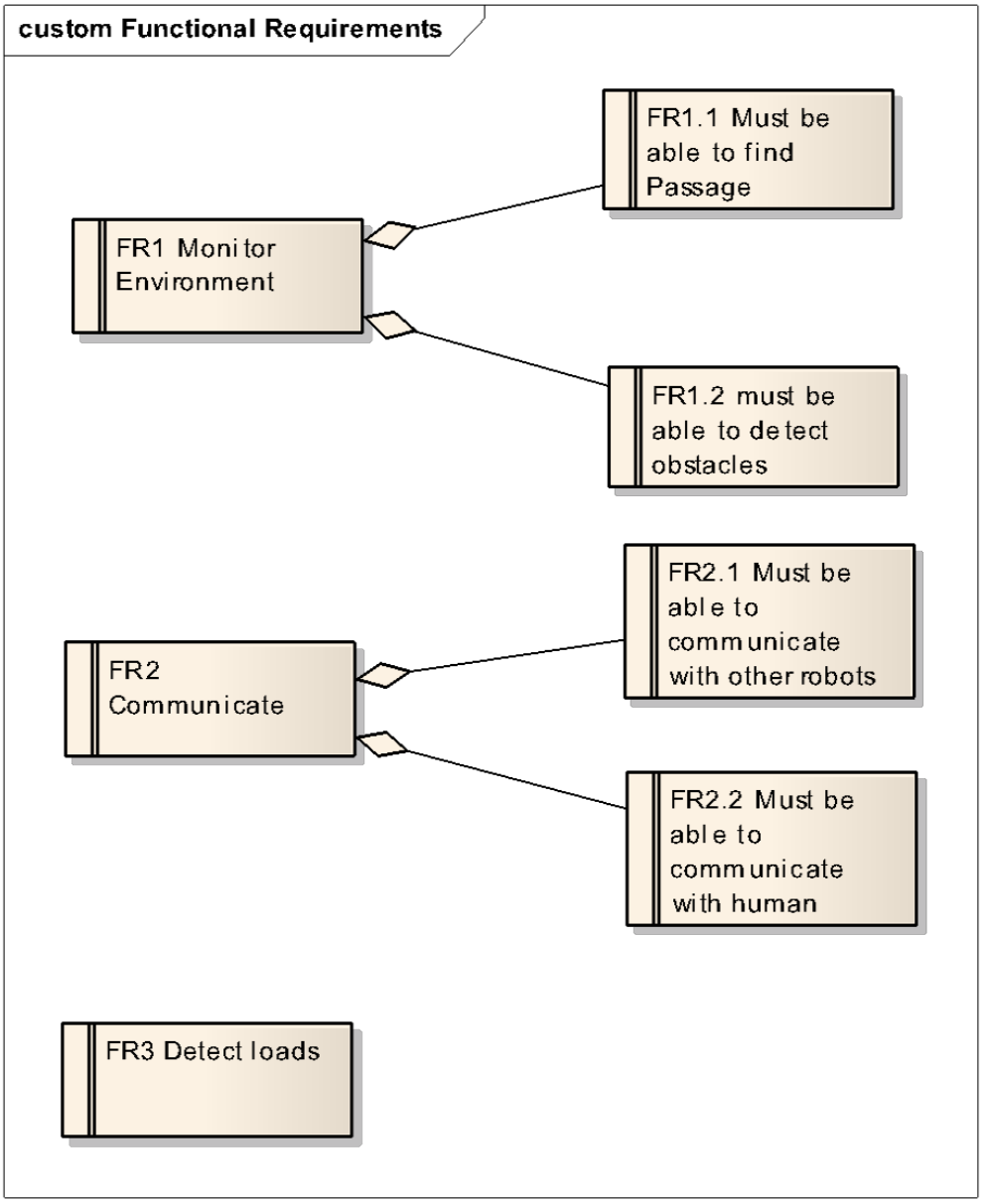
This use case is followed for the event of the robot monitors its environment for obstacles before traversing.

2.3 User Characteristics

The users of this system are the workers in a factory which conducts load. The general characteristic of the intended users is that they are individuals from lower educational level, they only experienced in conducting the robot and do not know how to maintain the robot.

3. SPECIFIC REQUIREMENTS

3.1 Software Product Features



3.1.1 Communicate with human Use Case

3.1.1.1 Brief Description

This use case is followed for the event of human communicates with the robot using LCD.

3.1.1.2 Flow of Events

Communicate with human - Basic Path

1. Human requests the information of the robots.
2. The embedded controller retrieves the data and display through a LCD.

3.1.1.3 Special Requirements

There are no special requirements associated with this use case.

3.1.1.4 Preconditions

The robots must be activated.

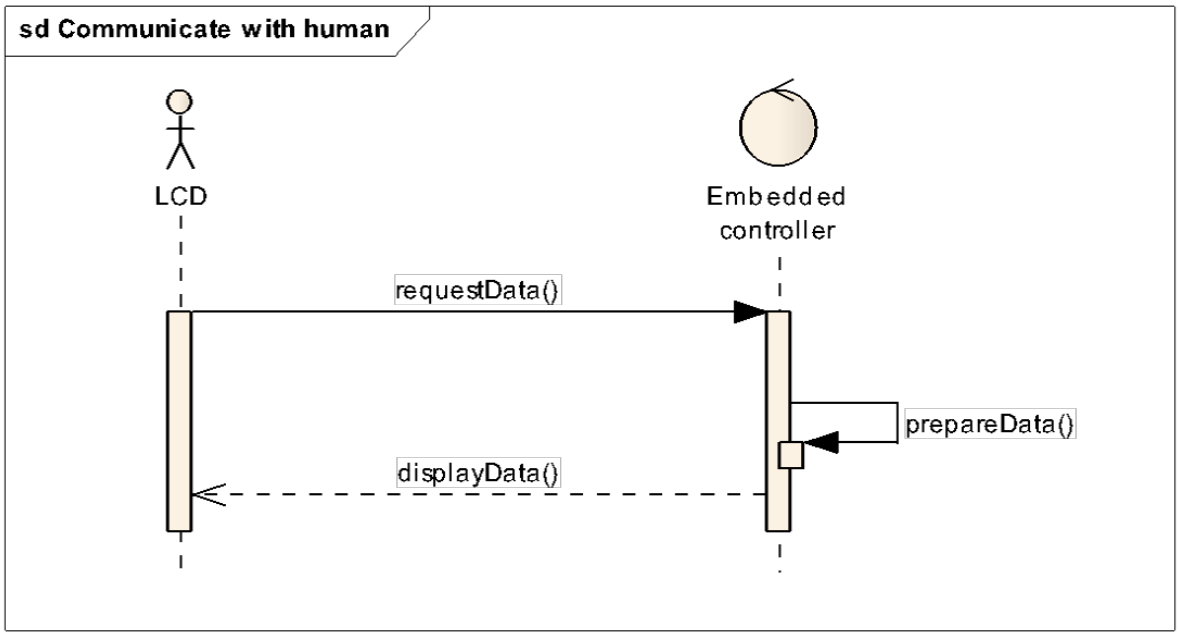
3.1.1.5 Postconditions

The human can communicate with the robots.

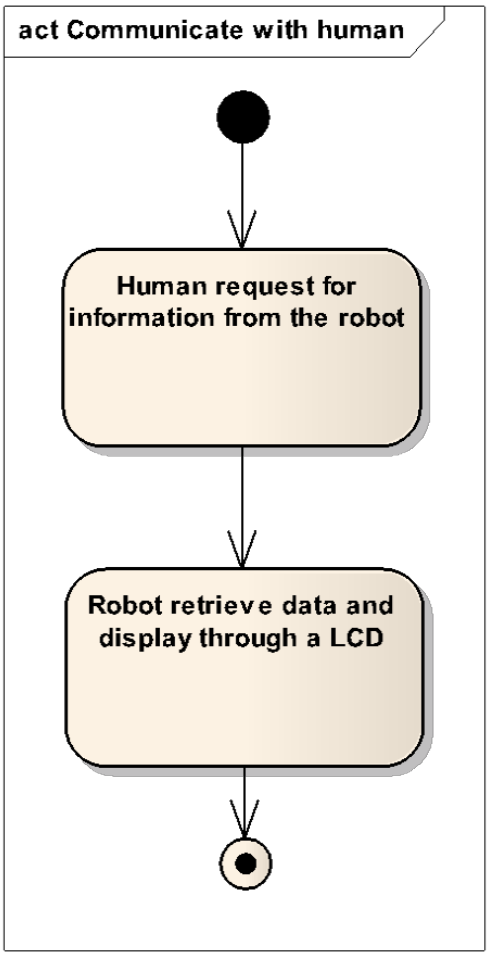
3.1.1.6 Extension Points

There is no extension points associated with this use case.

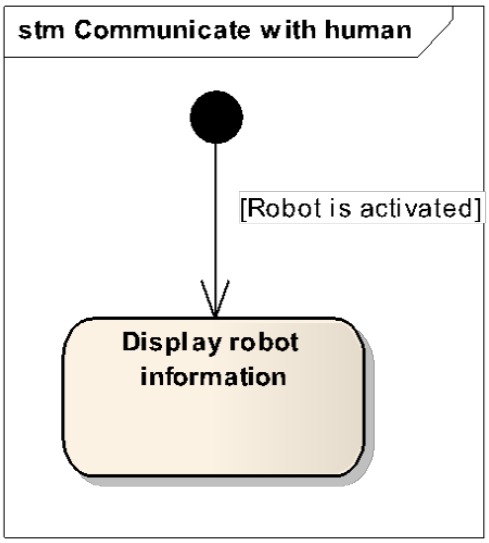
3.1.1.7 Communicate with human Sequence Diagram



3.1.1.8 Communicate with human Activity Diagram



3.1.1.9 Communicate with human Statechart Diagram



3.1.2 Communicate with other robots Use Case

3.1.2.1 Brief Description

This use case is followed for the event of the robot communicates with the other robots by sending message using Radio Frequency (RF) Transceiver.

3.1.2.2 Flow of Events

Communicate with other robots - Basic Path

1. Once the robots are activated and load is placed on the robots, it is ready to find passage.
2. Once the robots had detected the loads, it will send signal to the leader.
3. The robots will move according to the instruction that received from the leader. If the load had falls from the robots, the alternative flow will be follow.
4. The robots will finally reach the designated position.

Load falls - Alternate

3.1 Once the robots detect no load on top of it, it will stop moving and send a signal to

the leader.

3.2 After the leader had received the signal from the follower, it will signal the other followers to stop moving as well.

3.3 Once the load is placed back on top of the robot, the robot will send a signal to the leader.

3.4 The leader will signal all the followers to start moving and the flow control will back to 4.

3.1.2.3 Special Requirements

There are no special requirements associated with this use case.

3.1.2.4 Preconditions

The robot must be activated.

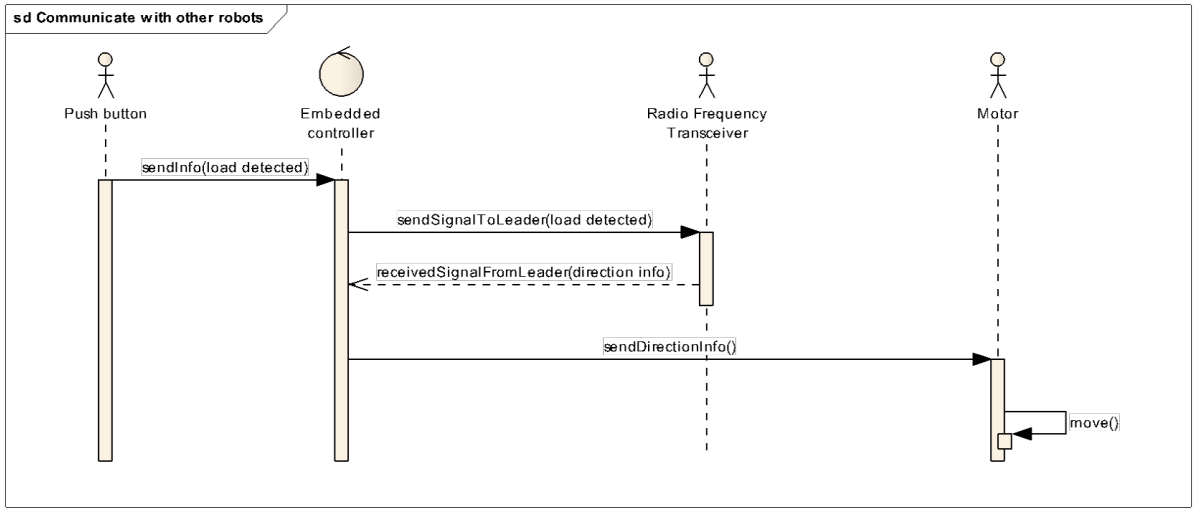
3.1.2.5 Postconditions

The robots can communicate with each others.

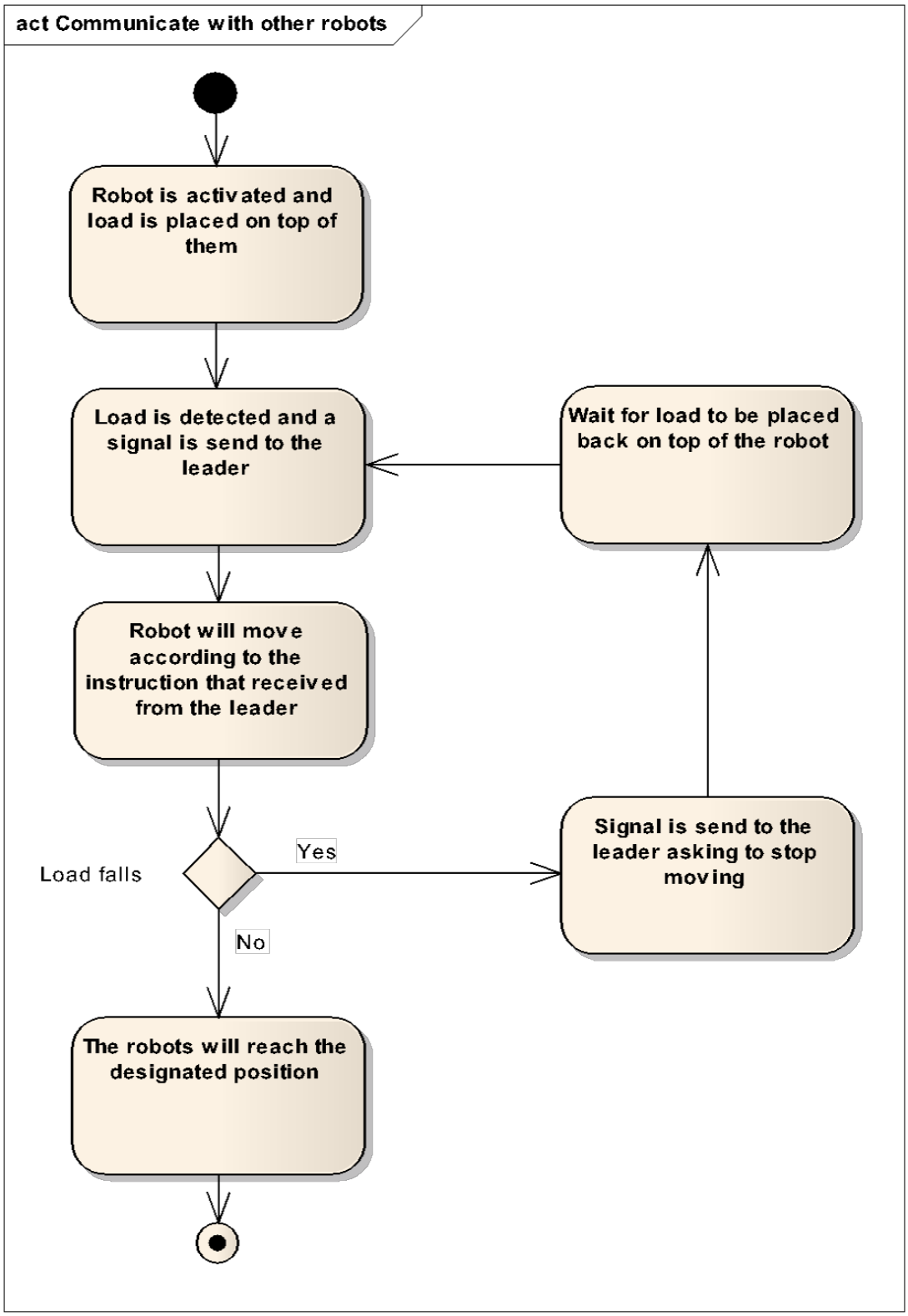
3.1.2.6 Extension Points

There is no extension points associated with this use case.

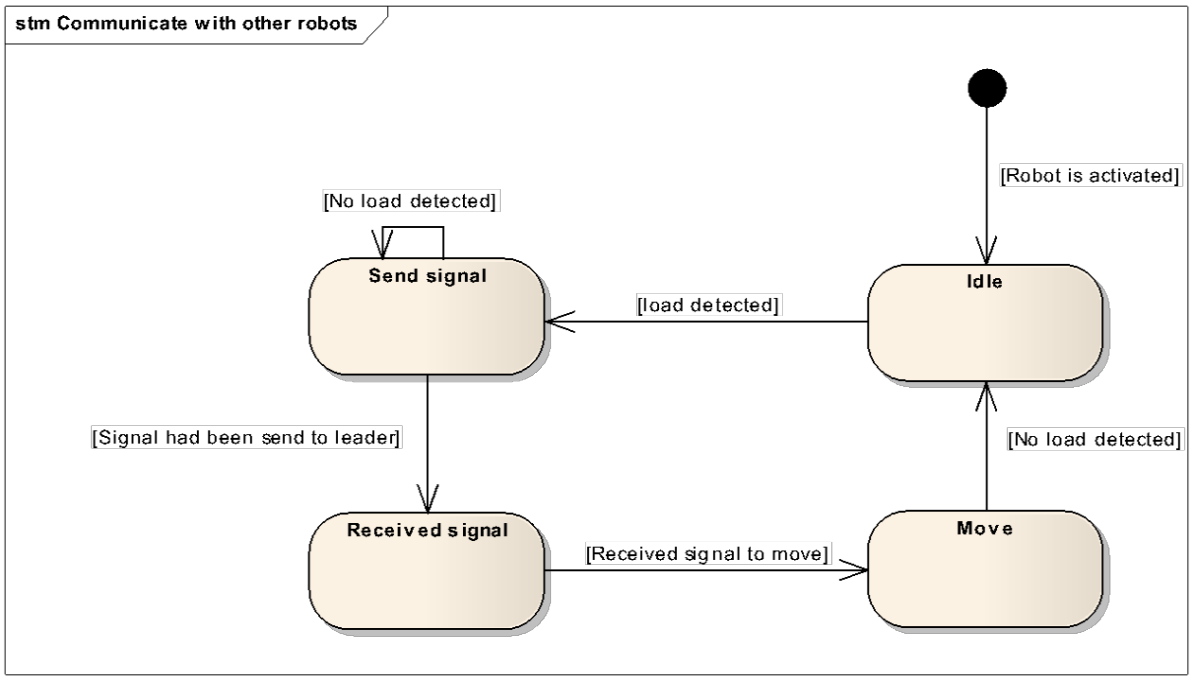
3.1.2.7 Communicate with other robots Sequence Diagram



3.1.2.8 Communicate with other robots Activity Diagram



3.1.2.9 Communicate with other robots State Chart Diagram



3.1.3 Detect loads Use Case

3.1.3.1 Brief Description

This use case is followed for the event of human placing a large object on top of the robot.

3.1.3.2 Flow of Events

Detect loads - Basic Path

1. A large object is being placed on the robot (on the push button) by a human.
2. The object presses the push button when it is being placed on top of the push button.
3. Once the push button is being pressed the weight of the object is sensed indicating the object has been placed on the robot already.
4. When the robot detects the load it is ready to traverse and find a passage. Once the loads fall, and the robot cannot detect any loads, alternative flow will be executed.
5. But the speed of the robot depends on the weight of the load: the heavier the load the slower the speed of the robot.

Loads fall - Alternate

4.1 If the load on the robot falls down when the robot is moving the push button pops up as there is no more weight on top of it.

4.2 Once the button pops up it is sensed that the load is not on the robot already.

4.3 A signal is sent to the robot to stop moving and the robot also communicates with the other follower or leader to stop as well.

4.4 The robot waits for human to put the load back on it.

4.5 Once the load has been replaced the robot communicates with the other robot and starts to move in their group once again.

3.1.3.3 Special Requirements

There are no special requirements associated with this use case.

3.1.3.4 Preconditions

The robot must be activated.

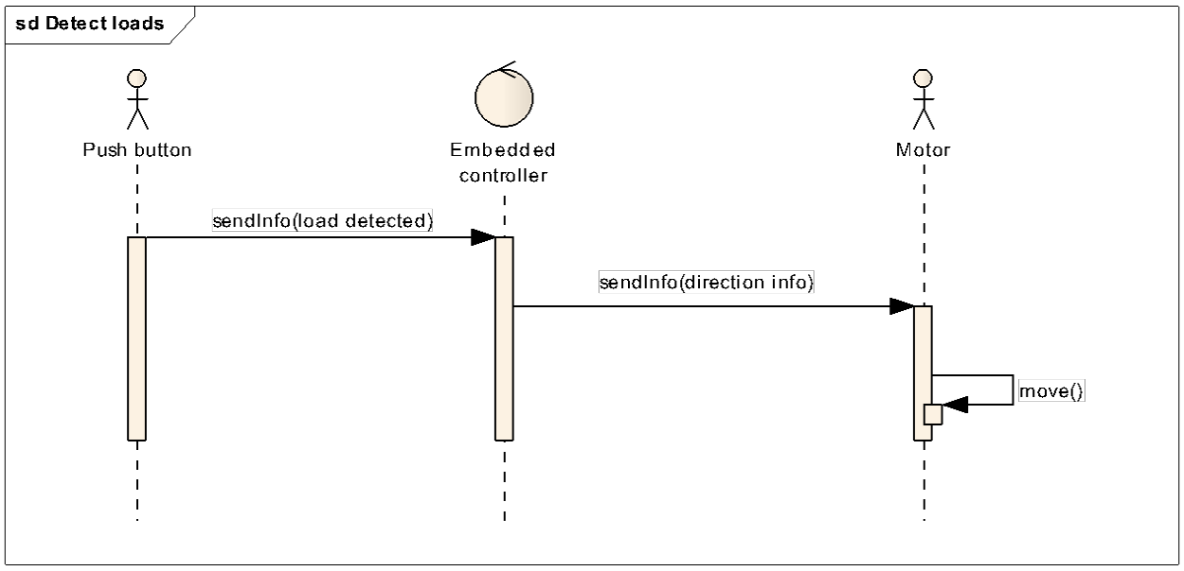
3.1.3.5 Postconditions

Loads are being placed on the robot.

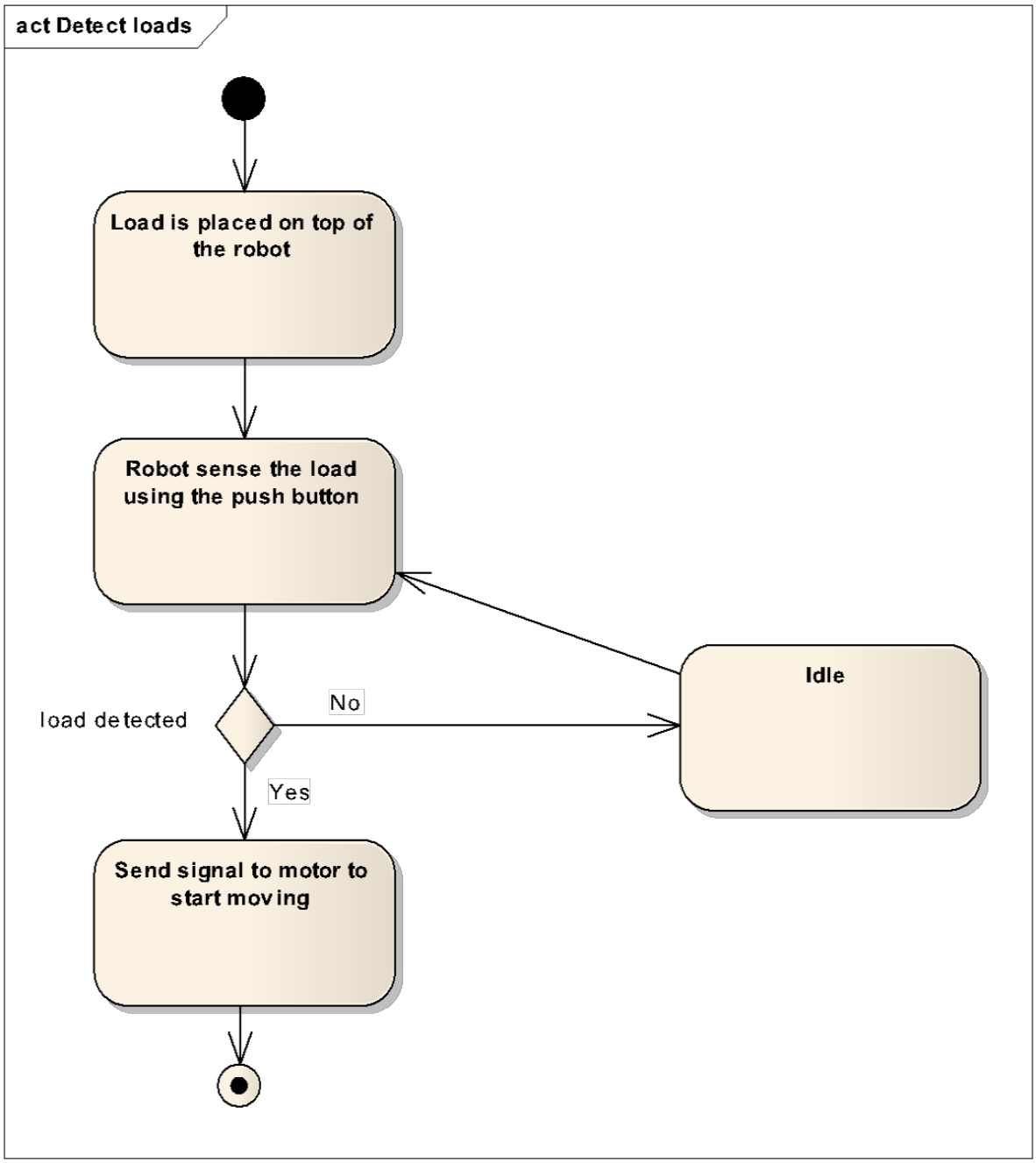
3.1.3.6 Extension Points

There is no extension points associated with this use case.

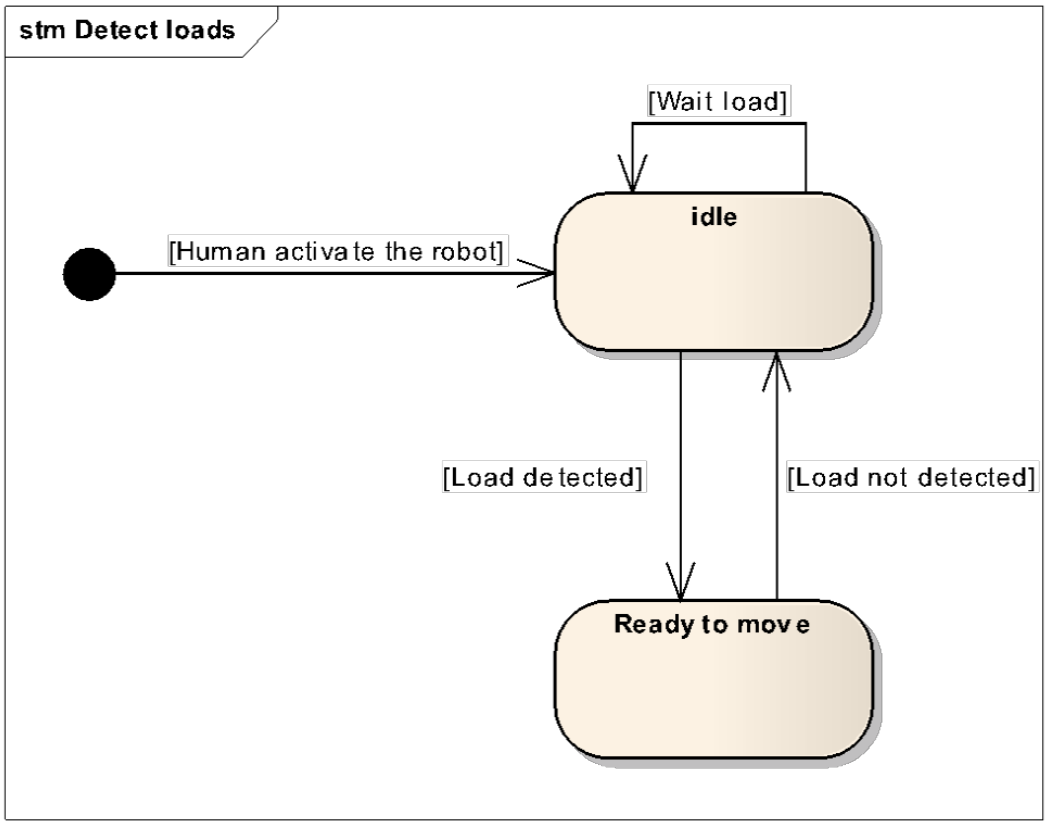
3.1.3.7 Detect loads Sequence Diagram



3.1.3.8 Detect loads Activity Diagram



3.1.3.9 Detect loads Statechart Diagram



3.1.4 Find passage Use Case

3.1.4.1 Brief Description

This use case is followed for the event of the robot moves to find a passage to deliver the load.

3.1.4.2 Flow of Events

Find passage - Basic Path

1. After load is placed on the robots and the motor is activated, robots start to move to designated position.
2. The robots will be able to find the passage according to the data they had received from the leader.
3. If the loads fall from the robots, alternative flows will be followed
4. When the robot reach the designated position, motor will stop and the session of

transportation is complete.

No loads detected - Alternate

3.1. When the robots detect no load on it, the robots will stop moving.

3.2. The robot will start moving when the load is put back on top of the push button and the flow control will be back to 4.

3.1.4.3 Special Requirements

There are no special requirements associated with this use case.

3.1.4.4 Preconditions

The robot must be activated.

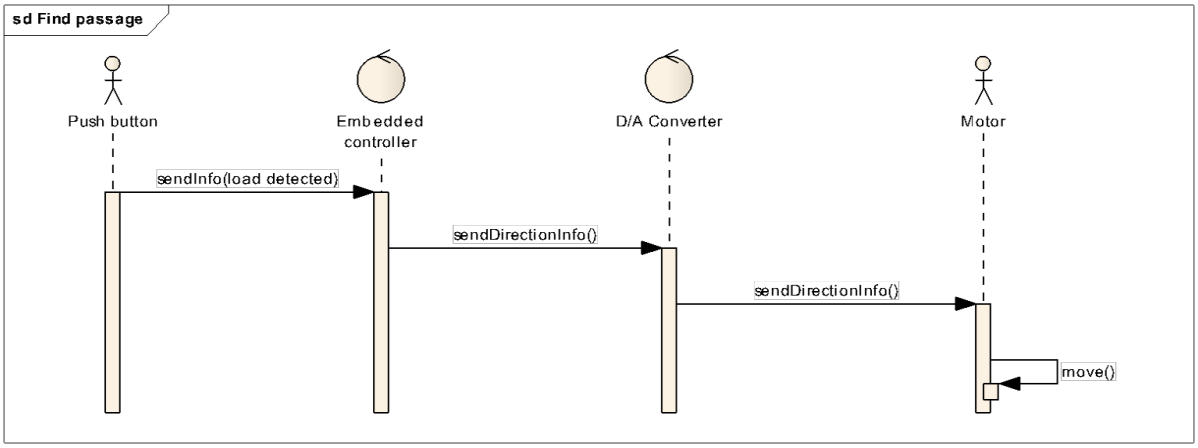
3.1.4.5 Postconditions

The robots are able to traverse and find passage.

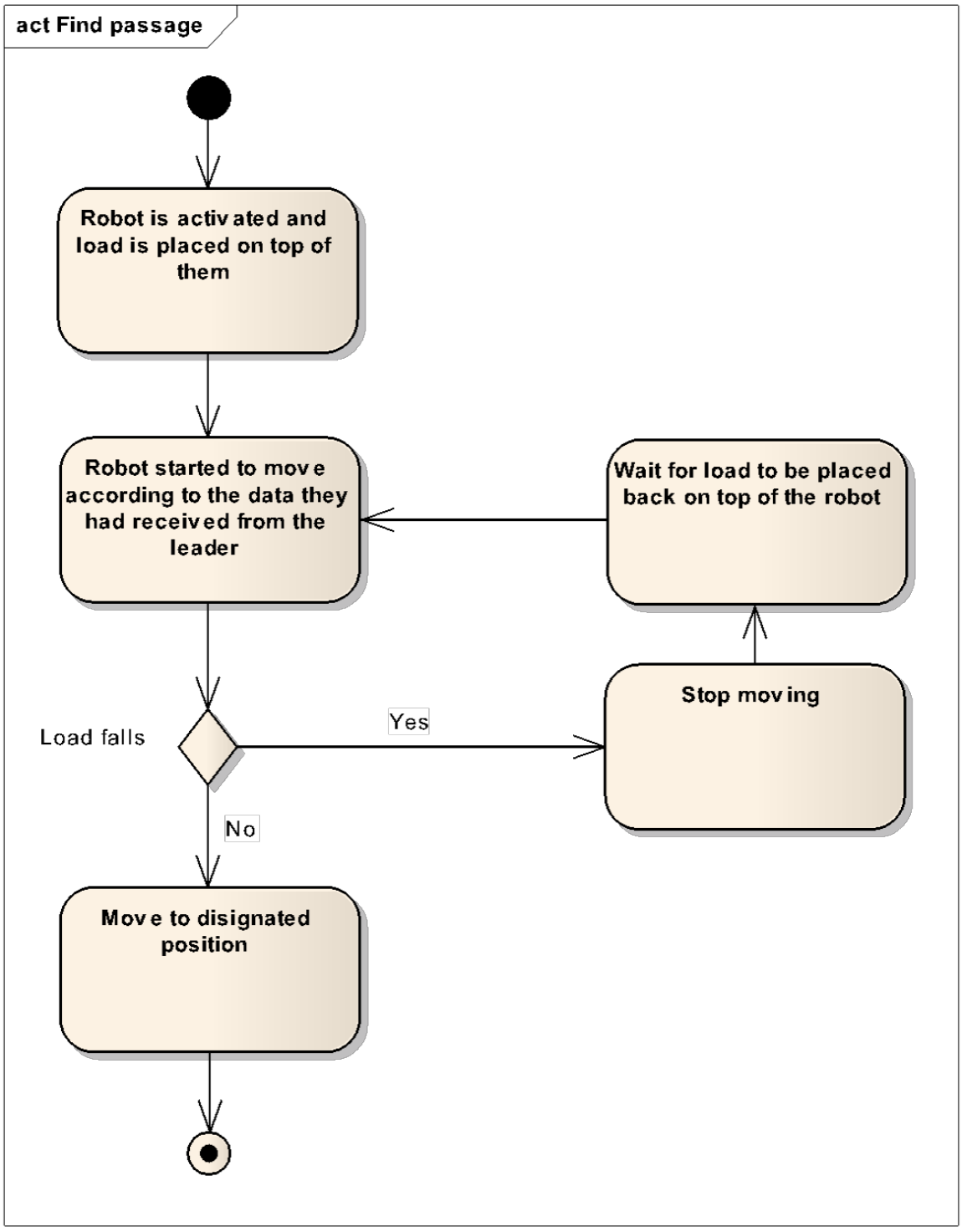
3.1.4.6 Extension Points

There is no extension points associated with this use case.

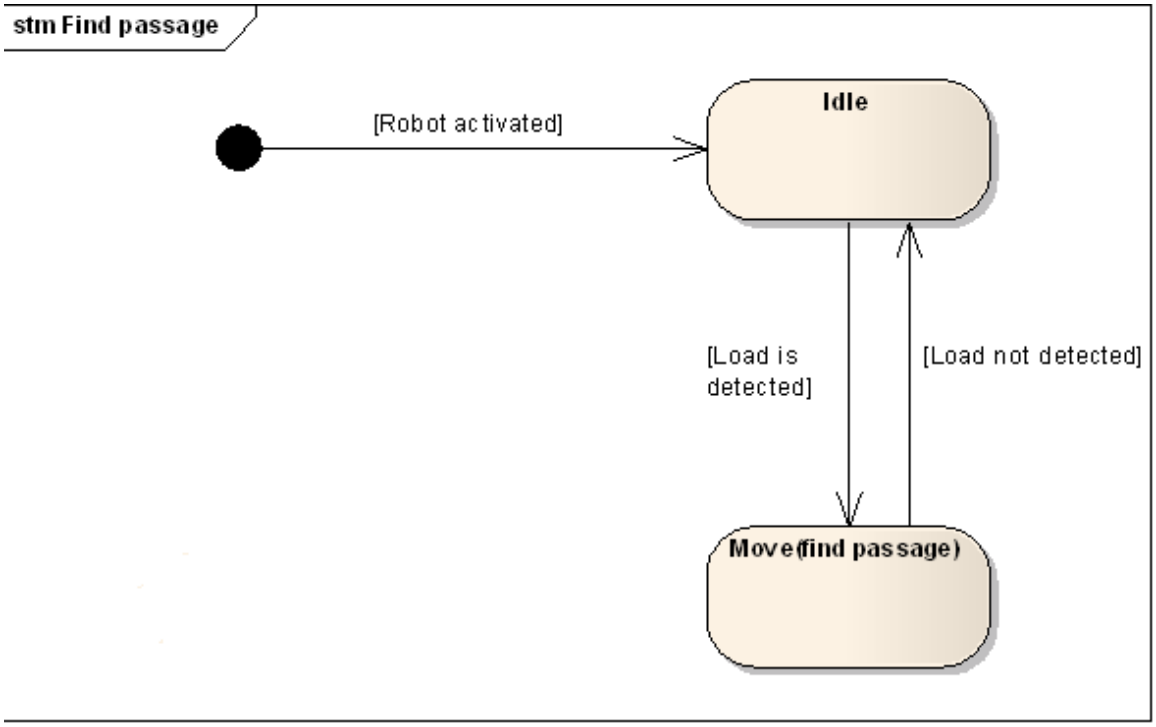
3.1.4.7 Detect loads Sequence Diagram



3.1.4.8 Find Passage Activity Diagram



3.1.4.9 Find Passage Statechart Diagram



3.1.5 Monitor environment Use Case

3.1.5.1 Brief Description

This use case is followed for the event of the robot monitors its environment for obstacles before traversing.

3.1.5.2 Flow of Events

Monitor Environment - Basic Path

1. Once the load is placed on the robots, it is ready to find passage for loading.

2. The embedded controller monitors its environment using the distance and proximity sensors.

2.1 When there are no obstacles, 3 are executed.

2.2 If obstacles detected, the alternative flow will be executed.

3. The robot moves according to the data they had received from the leader and will reach their designated position.

4. Once the robot has arrives their designated position, it will stop moving and waits for human to carry the loads.

Detect obstacles - Alternate

2.2.1. If an obstacle is detected, the robot will slows down its speed and the robot will not move towards the direction.

2.2.2. The embedded controller monitors the environment again for an area which free of obstacles.

2.2.3. The robot moves to the area which free of obstacles to avoid the obstacles.

2.2.4. Once the robot had avoided those obstacles, the flow control will back to 3.

3.1.5.3 Special Requirements

There are no special requirements associated with this use case.

3.1.5.4 Preconditions

The robot must be activated.

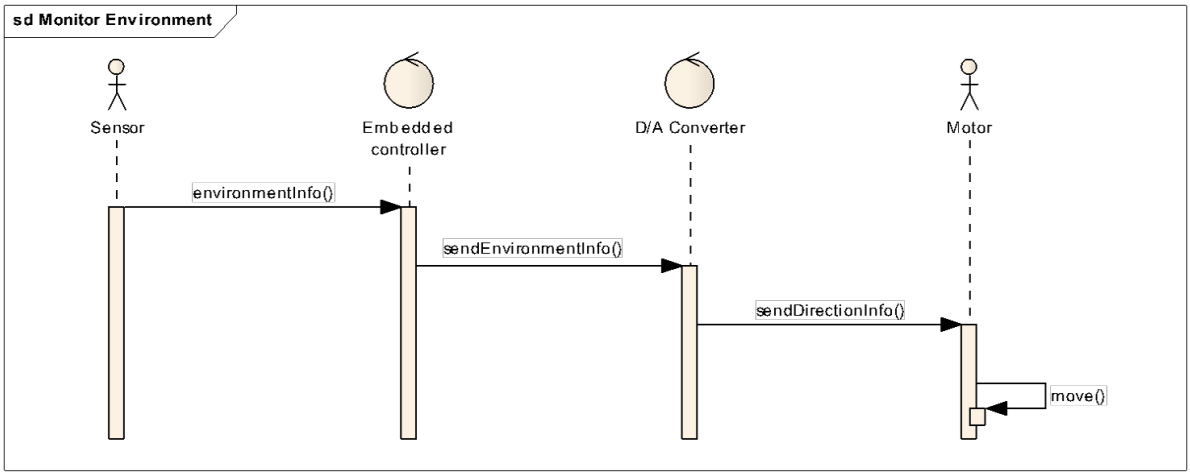
3.1.5.5 Postconditions

The robots are able to monitor the environment.

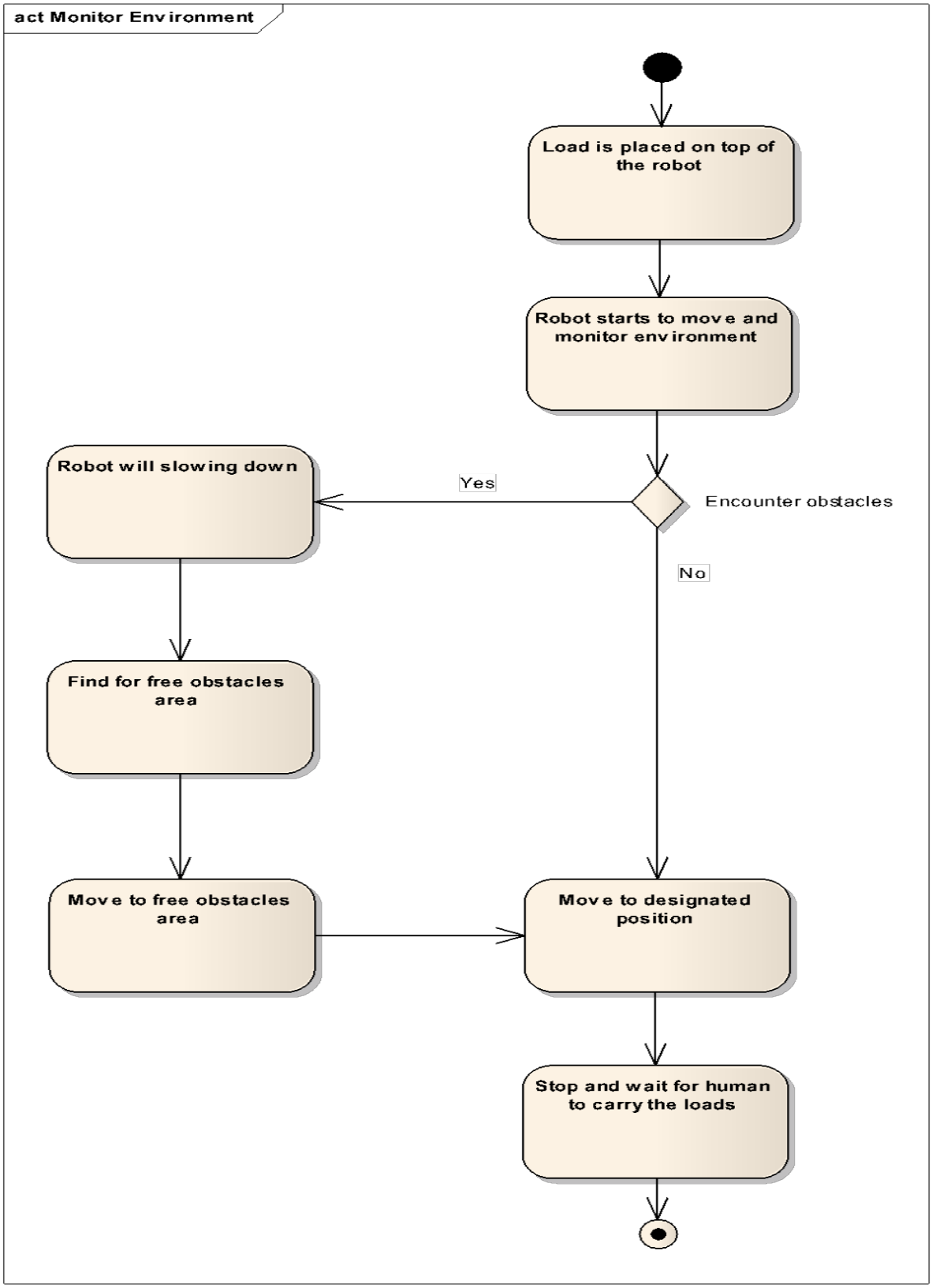
3.1.5.6 Extension Points

There is no extension points associated with this use case.

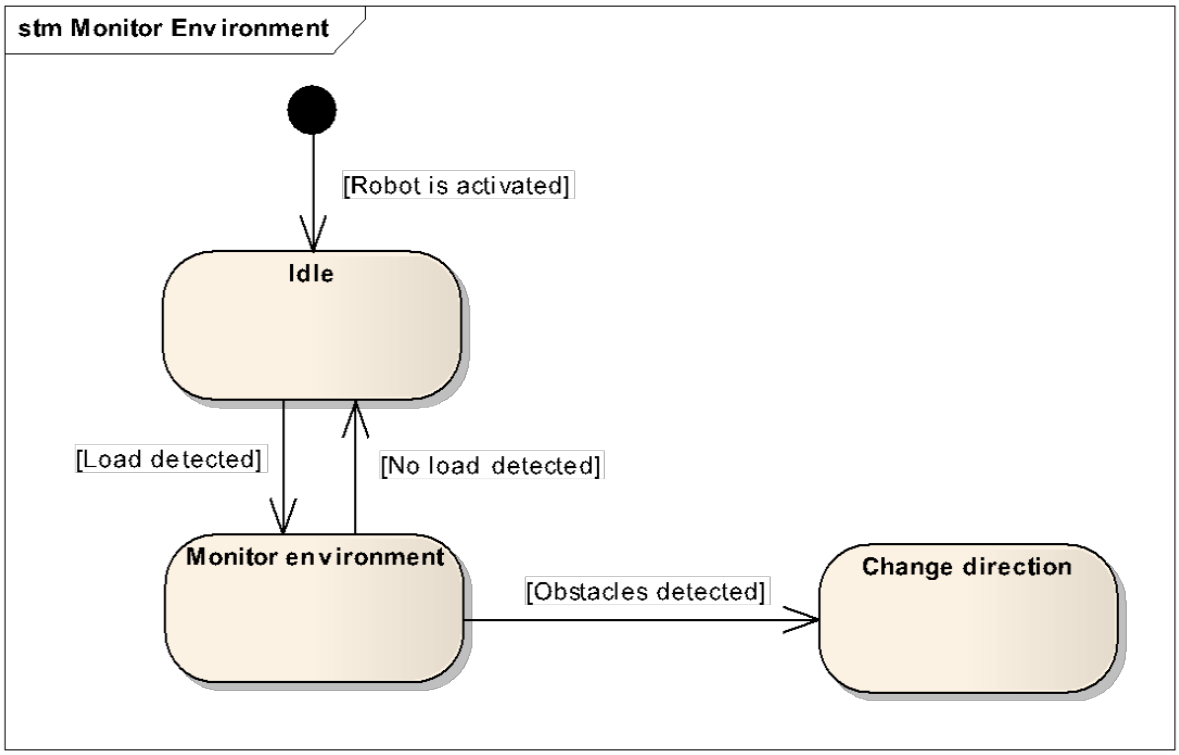
3.1.5.7 Monitor Environment Sequence Diagram



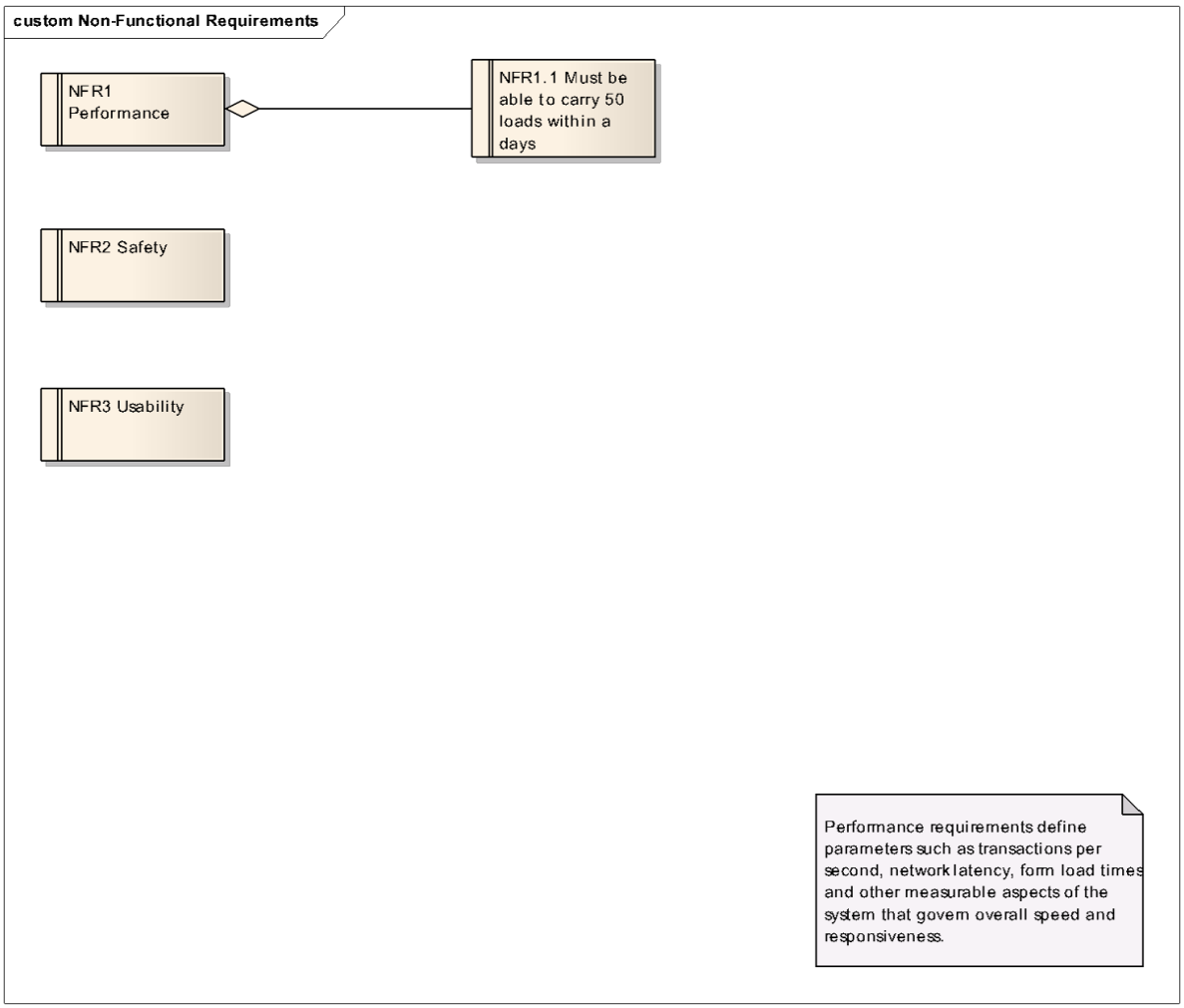
3.1.5.8 Monitor Environment Activity Diagram


act Monitor Environment
Load is placed on top of the robot
Robot starts to move and monitor environment
Encounter obstacles
Yes
No
Robot will slowing down
Find for free obstacles area
Move to free obstacles area
Move to designated position
Stop and wait for human to carry the loads

3.1.5.9 Monitor Environment Statechart Diagram



3.2 Non Functional Requirements



3.2.1 Usability

The load transportation robot system must be designed for ease-of-use. Furthermore, a reference of the load transportation robot system must have built-in to the system to help the user

3.2.2 Safety

The physical design of the load transportation robot must be well design and will not cause any danger to human when accident occurred such as short circuit.

3.2.3 Performance

The load transportation robot must be able to carry 50 loads within a day.

4. REQUIREMENTS TRACEABILITY

